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Dept: Data engineering

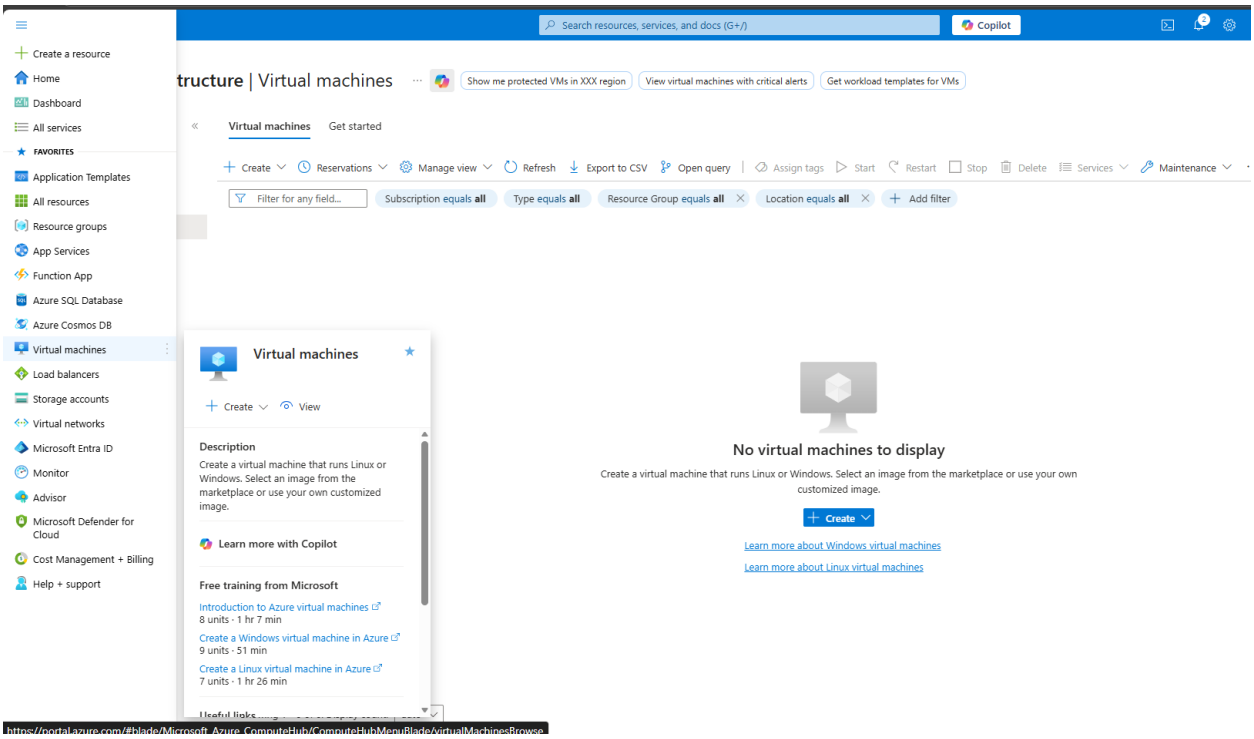
Tasks:

1. Create an Azure Virtual Machine
2. Start, Stop, and Restart a Virtual Machine
3. Add a managed disk to the VM.

Creation of azure VM

Click on 3 lined option on top left corner of azure portal to open left side bar

Click on azure virtual machines



Click on create VM:-

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimized for high a

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. Learn more

This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * : Azure for Students

Resource group * : (New) vm1_group

Create new

Instance details

Virtual machine name * : vm1

Region * : (Asia Pacific) Central India

Deploy to an Azure Extended Zone

Availability options : No infrastructure redundancy required

Security type : Standard

Image * : Ubuntu Server 24.04 LTS - x64 Gen2

See all images | Configure VM generation

This image is compatible with additional security features. Click here to swap to the Trusted launch security type.

VM architecture :

- ○ Arm64
- ◉ x64

VM architecture

- ○ Arm64
- ◉ x64

Run with Azure Spot discount ☐

Size * Standard_B2ats_v2 - 2 vcpus, 1 GiB memory ($4.49) (free services eligible)

See all sizes

Enable Hibernation ☐

Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernate to enable this feature. Learn more

Administrator account

Authentication type

- ◉ SSH public key
- ○ Password

Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.

Username * azureuser

SSH public key source Generate new key pair

SSH Key Type

- ◉ RSA SSH Format
- ○ Ed25519 SSH Format

Ed25519 provides a fixed security level of no more than 128 bits for 256-bit key, while RSA could offer better security with keys longer than 3072 bits.

Key pair name * vm1_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

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Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports *

- ○ None
- ◉ Allow selected ports

Select inbound ports * SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Disk settings:

Create a virtual machine ··· Help me choose the right VM size for my workload | Help me create a VM optimized for high availat

Basics | **Disks** | Networking | Management | Monitoring | Advanced | Tags | Review + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. Learn more

There is a charge for the underlying storage resources consumed by your virtual machine. Learn more

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host ☐

Encryption at host is not registered for the selected subscription. Learn more

OS disk

OS disk size: Image default (30 GiB)

OS disk type *: Premium SSD (locally-redundant storage)

Delete with VM ☑

Key management: Platform-managed key

Enable Ultra Disk compatibility ☐

Ultra disk is supported in Availability Zone(s) 2,3 for the selected VM size Standard_B2ats_v2.

Data disks for vm1

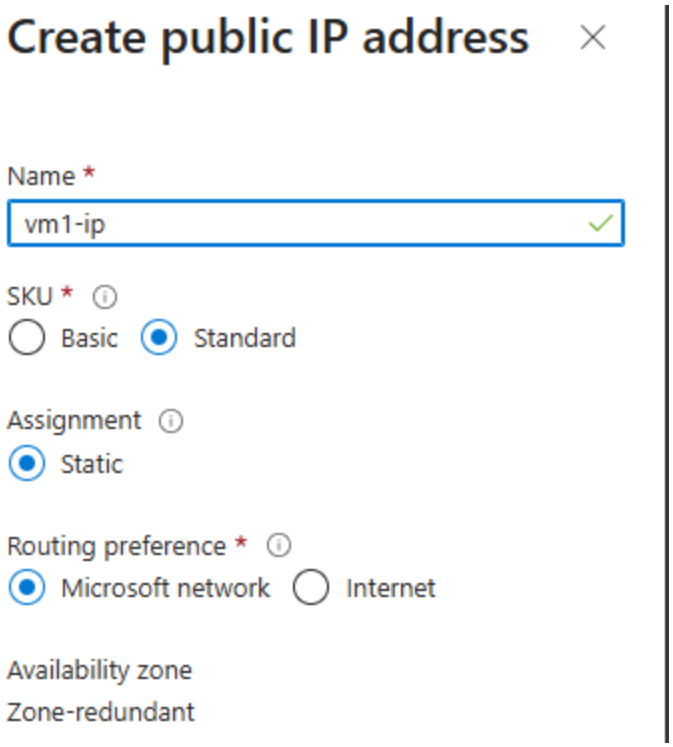
You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching	Delete with VM

Create and attach a new disk Attach an existing disk

Network settings:

Creating public IP:



Imp: select auto delete of NIC and public IP after VM deletion for saving cost

Create a virtual machine

Help me choose the right VM size for my workload | Help me create a VM optimiz

Basics | Disks | **Networking** | Management | Monitoring | Advanced | Tags | Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *
(new) vm1-vnet
Create new

Subnet *
(new) default (10.0.0.0/24)

Public IP
(new) vm1-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group
- None
- Basic
- Advanced

Public inbound ports *
- None
- Allow selected ports

Select inbound ports *
SSH (22)

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Delete public IP and NIC when VM is deleted [x]

Enable accelerated networking [ ]

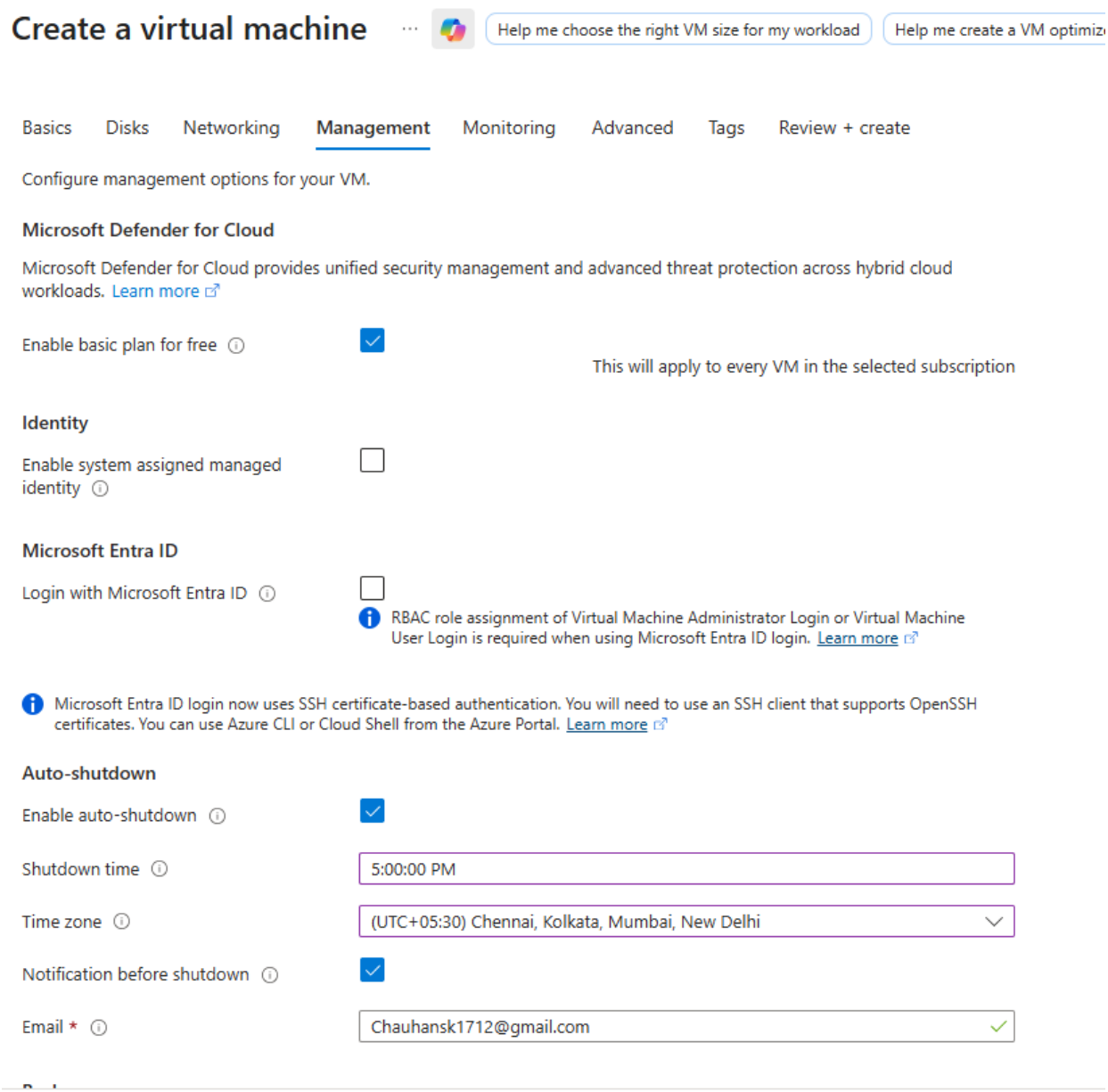
Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. Learn more

Load balancing options
- None
- Azure load balancer
  Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.
- Application gateway
  Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

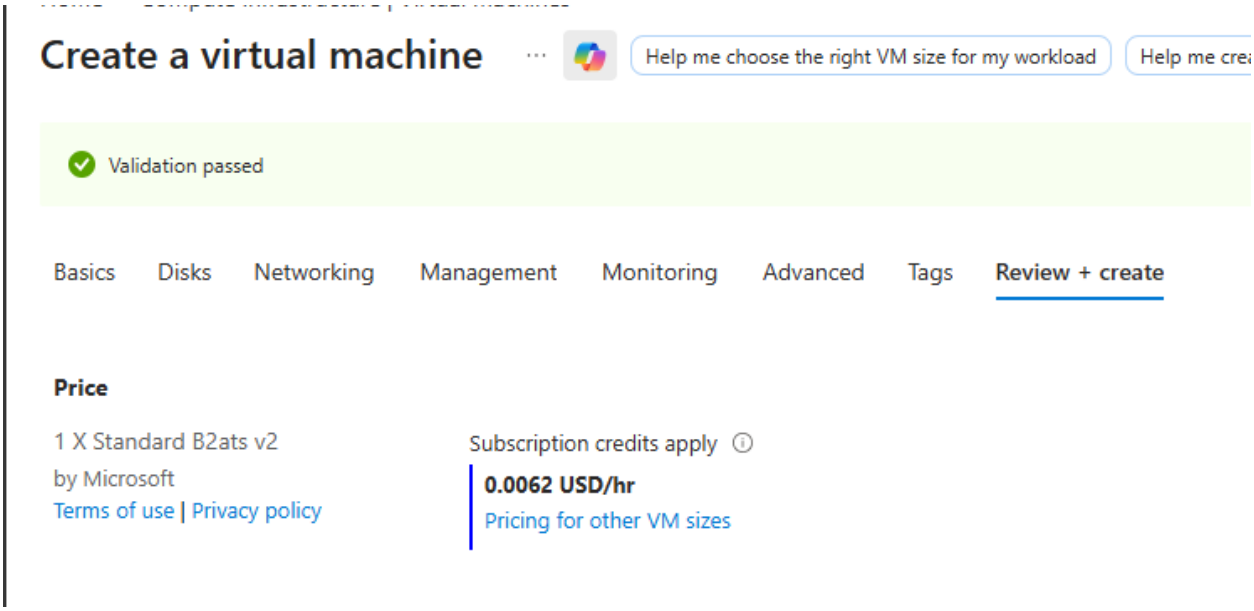
In the management tab:

Configure auto-shutdown for saving cost



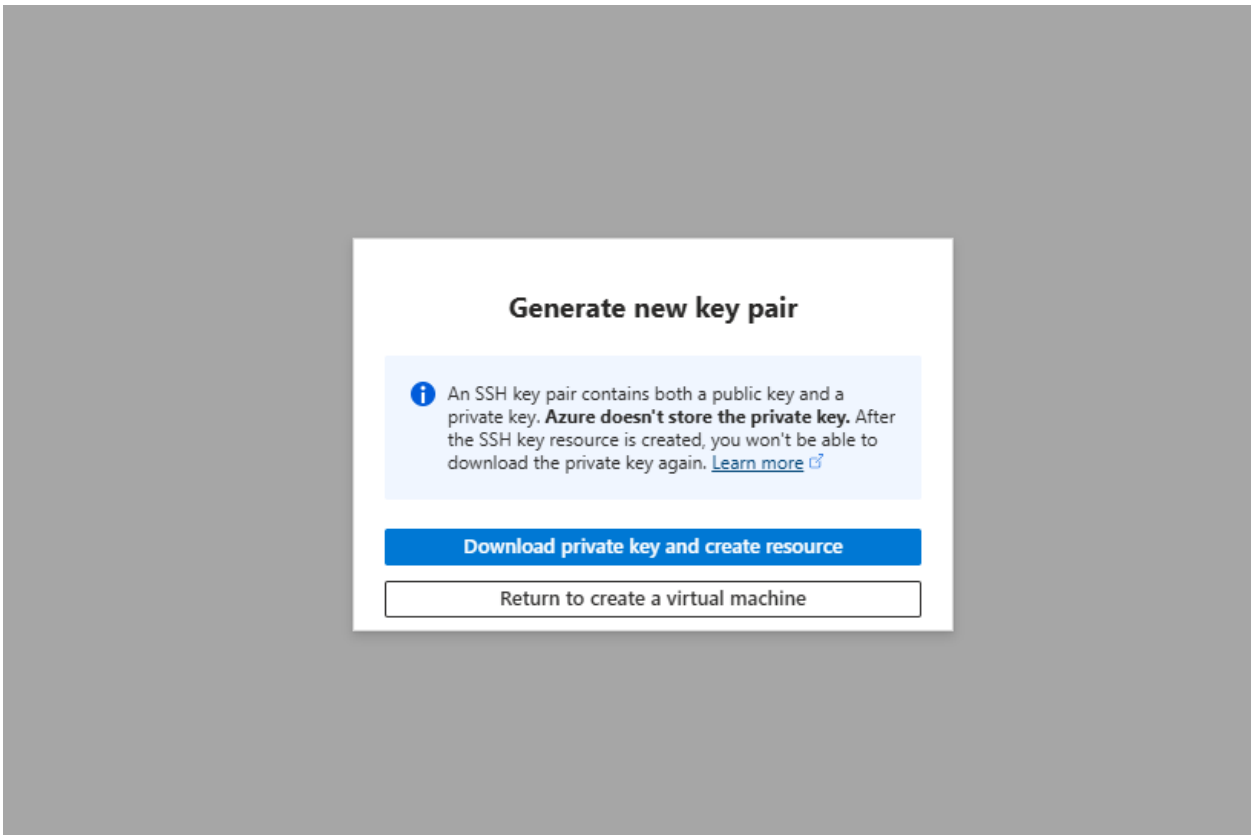
Skiping other settings to default

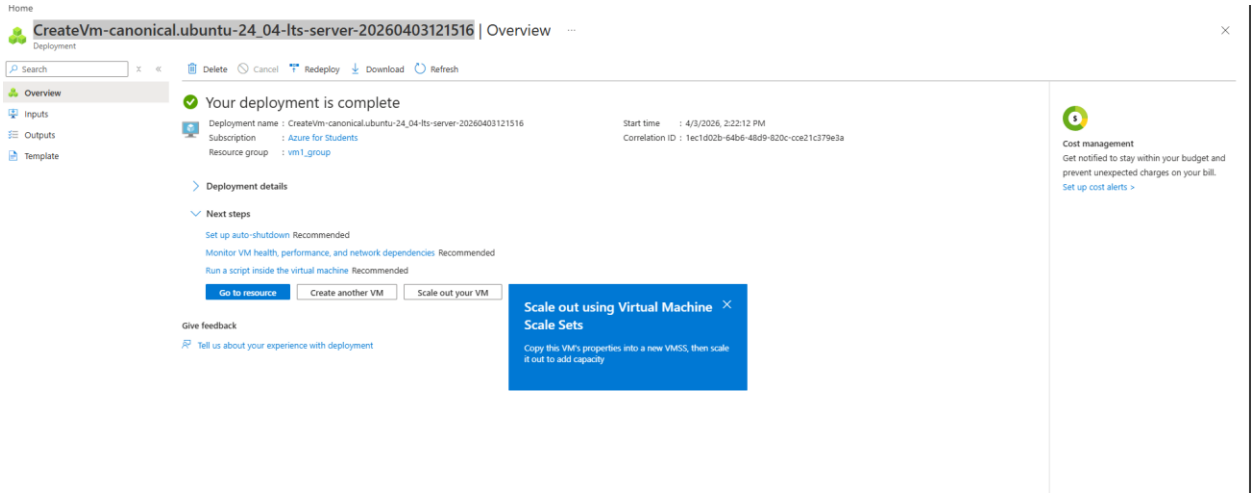
In review + create tab you will be able to see the final cost/ hr



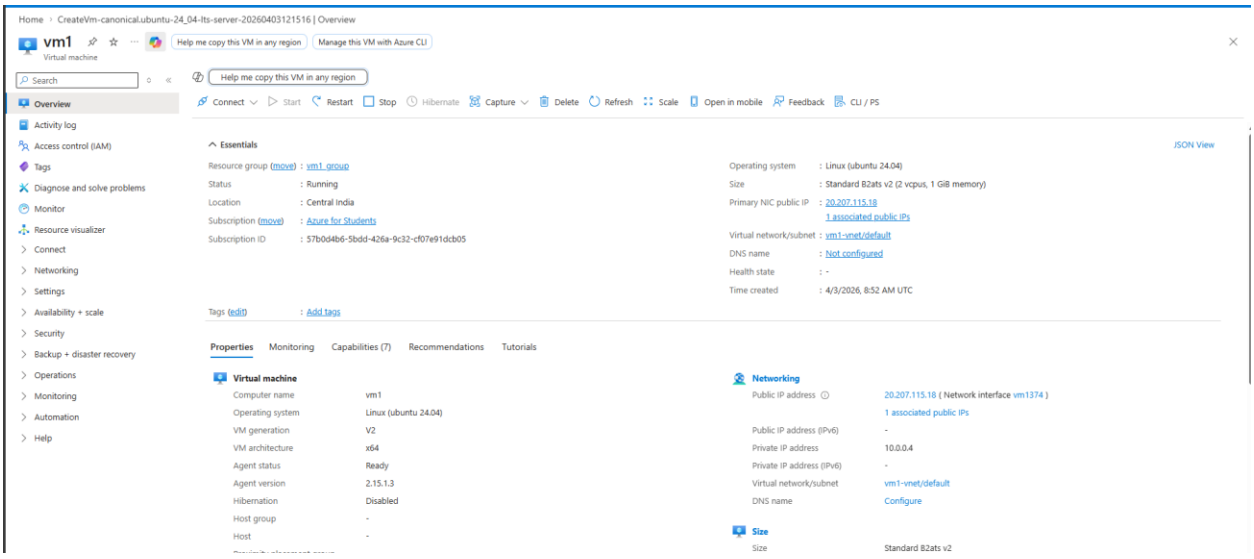
After hitting the create button:

You will ask to download private key, you must download it otherwise you will not be able to download it later

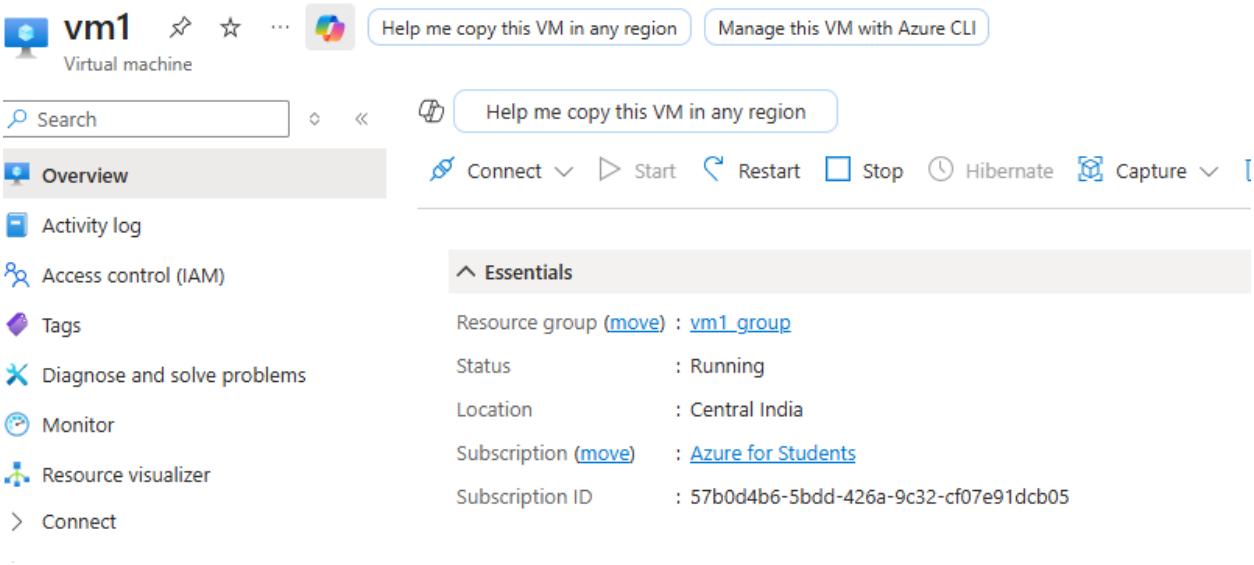




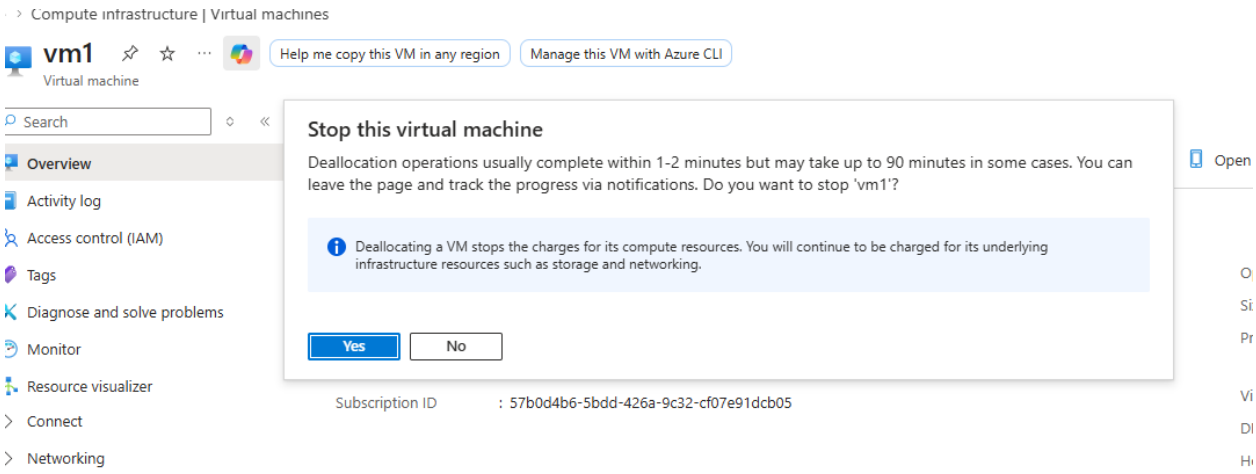
Click on go to resources:



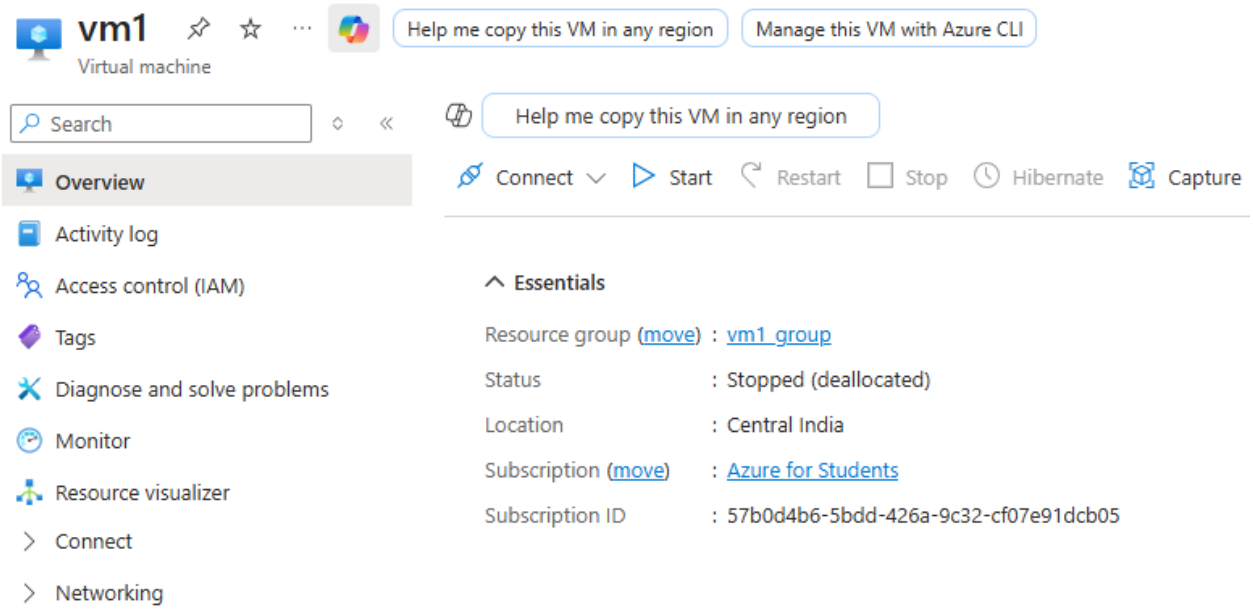
Status: running



Stopping the VM:



Status: stop



We can restart by clicking on restart button

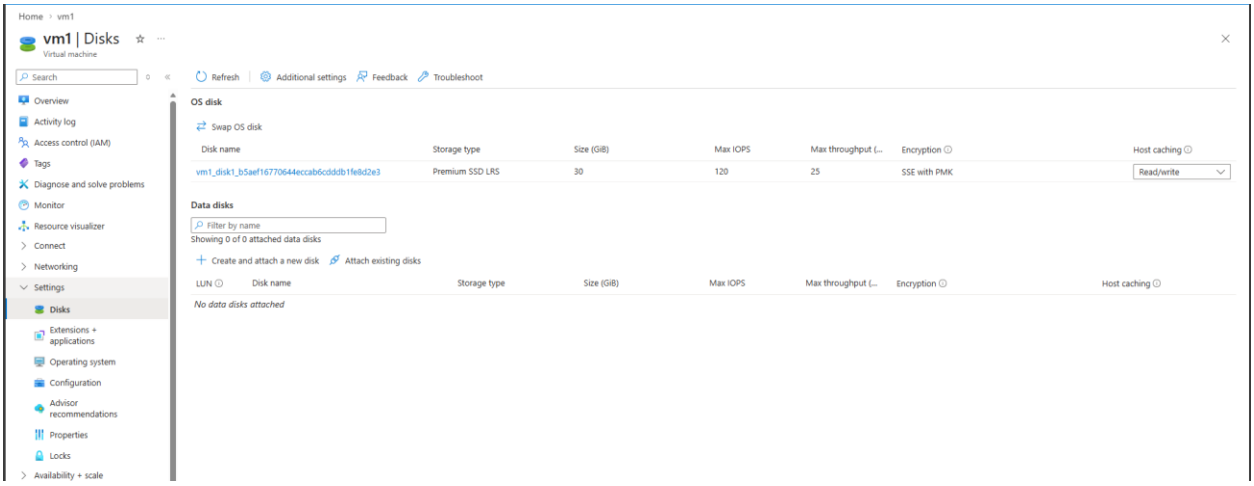
It will appear once you start your VM

Attaching data disk

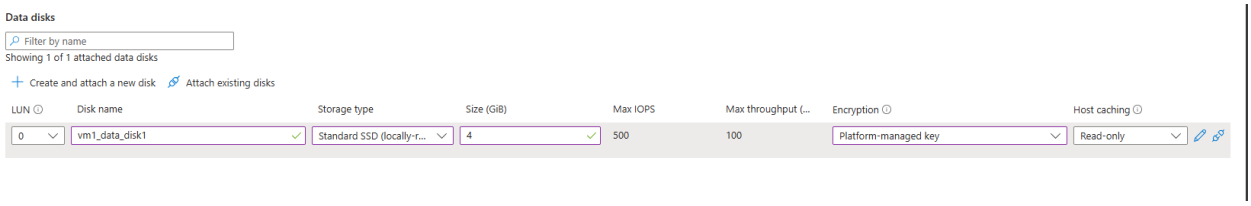
for attaching data disk you need to open disk settings in Vm1

In the below image you can see OS disk and data disks

Create and attach one data disk to it

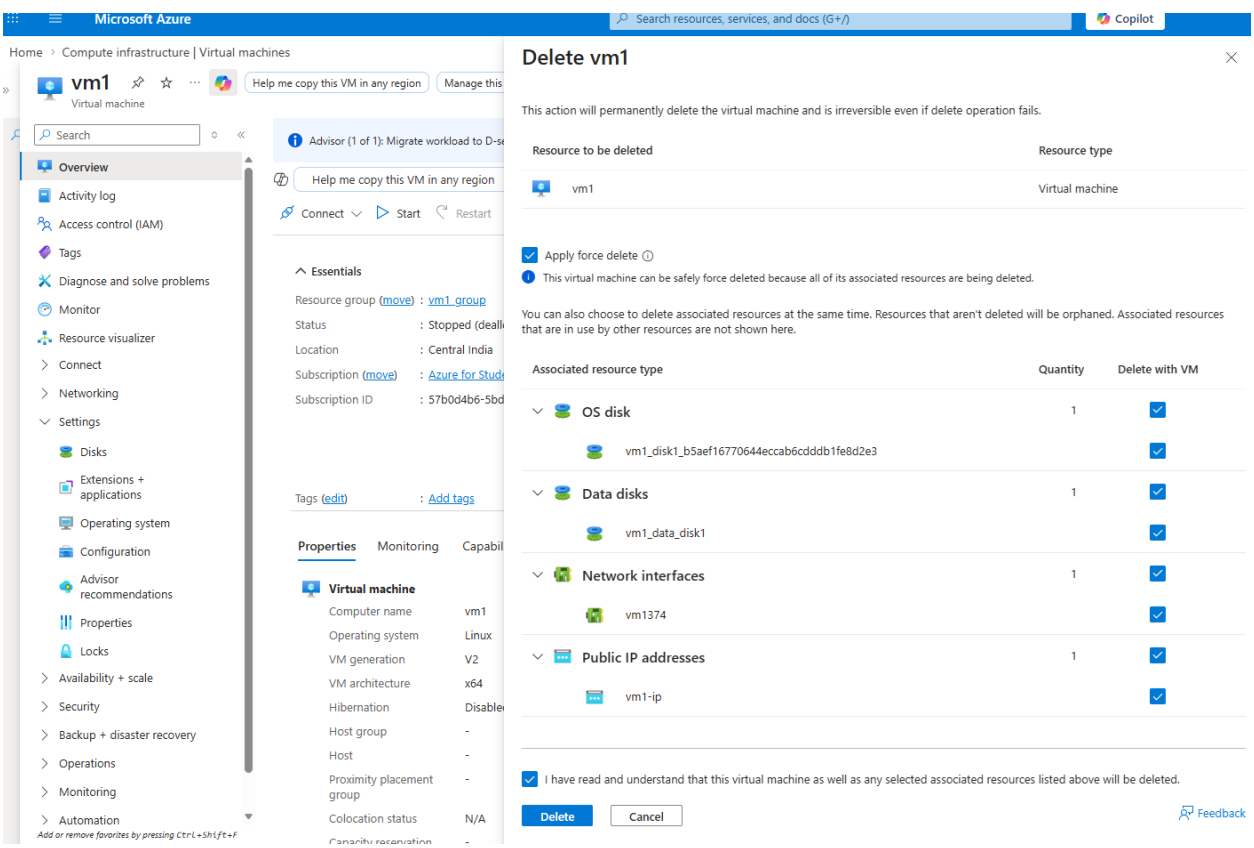


Adding new data disk



Connecting vm using SSH on windows:

You need to have public ip of your

```
azureuser@vm1: ~

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\system32> icacls "C:\Users\sumit.chauhan\Downloads\vm1_key.pem" /inheritance:r /grant:r "$($env:USERNAME):
(R)"
processed file: C:\Users\sumit.chauhan\Downloads\vm1_key.pem
Successfully processed 1 files; Failed processing 0 files
PS C:\Windows\system32> ssh -i "C:\Users\sumit.chauhan\Downloads\vm1_key.pem" azureuser@20.207.115.18
The authenticity of host '20.207.115.18 (20.207.115.18)' can't be established.
ED25519 key fingerprint is SHA256:FaVyN2LFELCyeg+xCmBtjjoMFyMpJrPMt5DtN0CSP5M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.207.115.18' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1010-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr  3 09:29:14 UTC 2026

  System load:  0.0               Processes:             118
  Usage of /:   5.8% of 28.02GB   Users logged in:       0
  Memory usage: 33%               IPv4 address for eth0: 10.0.0.4
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.
```

Deleting VM: